03. MESODERM AND NOTOCHORD FORMATION

DURATION : 03:43 STUDY SHEET

COURSE SUMMARY

This video guides you through the fascinating process of mesoderm formation and the crucial role of the notochord in embryonic development. You will learn how these structures interact at a cellular level, particularly through the primitive streak and bottle cells, leading to the formation of primitive mesenchymal tissue. Furthermore, you will explore practical aspects related to the coccyx, discovering how impulses and electromagnetic fields influence tissue health. The session includes a practical approach to notochordal movements while highlighting the interconnection between the sacrum, coccyx, and even the heart, thus deepening your understanding of biodynamic embryology and osteopathy.

ESTABLISHMENT OF THE MESODERM AND NOTOCHORD

THE ESTABLISHMENT OF THE MESODERM

The first crucial step is the **establishment of the mesoderm**. We observe the appearance of the **primitive streak** and the **notochordal axis**.

There is an aspiration field between the **ectodermal** tissue and the **endodermal** tissue, creating a movement where a space will be filled.

Mesodermal tissue originates from **epiblastic** tissue, not ectodermal tissue. The **bottle cells** are drawn towards the centre.

This is how the space between the epiblastic and hypoblastic tissue becomes the **primitive mesenchymal tissue**, the future mesoderm.

ROLE OF THE NOTOCHORD

The **notochord** influences this process. It is an electrical field that acts.

In the previous lesson, we worked at the level of the **sacrum**, focusing on the notochordal axis.

We will explore the process in more depth. The notochord, positioned here, is linked to **Hensen's node**.

This generates a wave-like movement, a **cranio-caudal** movement, where the primitive streak progressively regresses backwards.

This primitive streak is found at the **sacrococcygeal** base, constituting an energetic vestige.

Thus, the establishment of the mesoderm is observed.

THE PRACTICE: THE COCCYX AND IMPULSES

We will practice by focusing on the **coccyx**, as this area concentrates energy.

The first impulse occurs on the **hypoblastic** tissue. It is at this moment that the epiblast becomes **ectoderm** and the hypoblast becomes **endoderm**.

This means that the epiblast informs all my tissues, and it is itself completely integrated by my notochord. It is an **electromagnetic** field that informs the body.

When we consciously touch this level, we are connected to this force that the body has established.

DEVELOPMENT OF MESODERMAL TISSUE

We arrive at the second impulse, the second wave, which will form the **mesodermal tissue**.

I thus obtain an initial notochordal information which, with my primitive streak, generates this aspiration field.

The latter creates the space of my primitive streak, in contact with the establishment of my **embryonic pedicle** and my fields of **permeation** and **infusion** that are organising themselves.

It is in this movement that the integration of my mesoderm takes place. This occurs between the **fourteenth** and **sixteenth day**.

When you practice, you first perceive this kind of **"Tai Chi"**, this notochordal wave.

Then, underneath, at the same time as this movement, there is an invasion underneath. It is a **double movement**.

One can position oneself on the sacrum or on the coccyx. Subsequently, we will address other very specific balance points at the level of the **heart**.

The presumptive aspect of the heart is determined by this field of action.

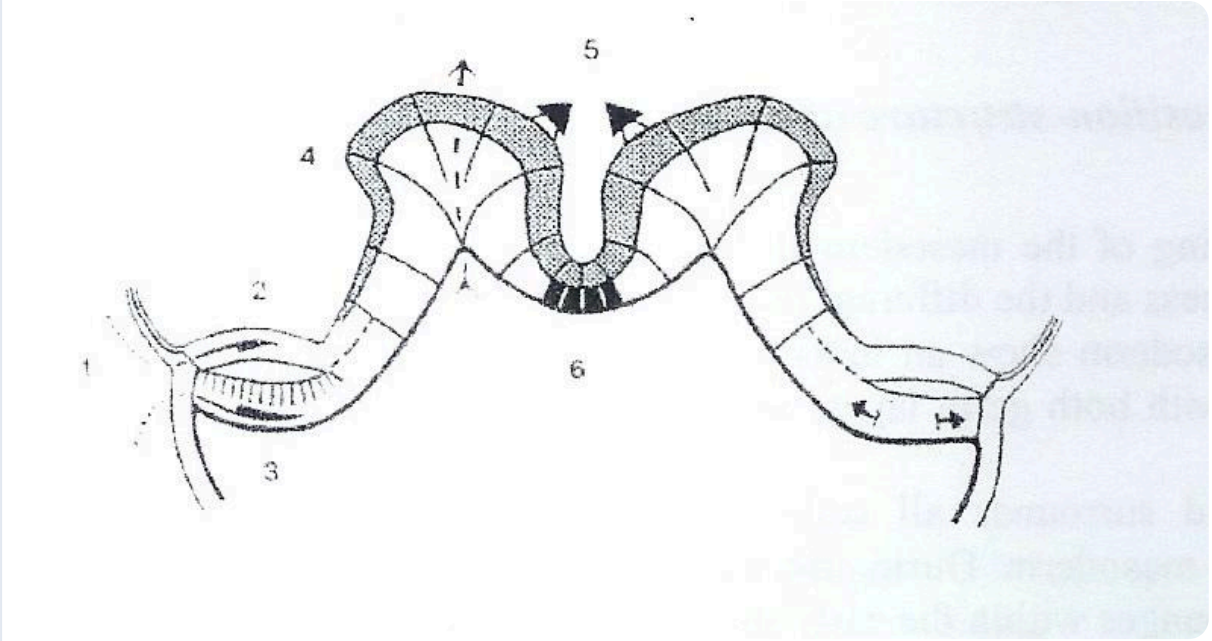


FIG. — Beginning of neurulation

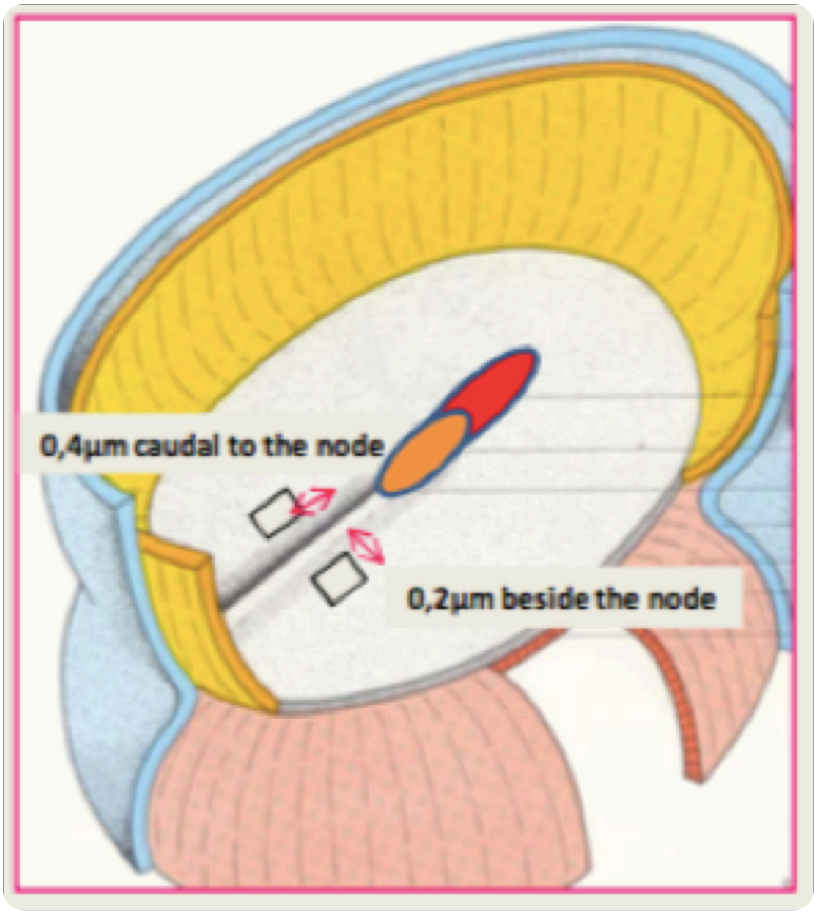


FIG. — Early embryonic development

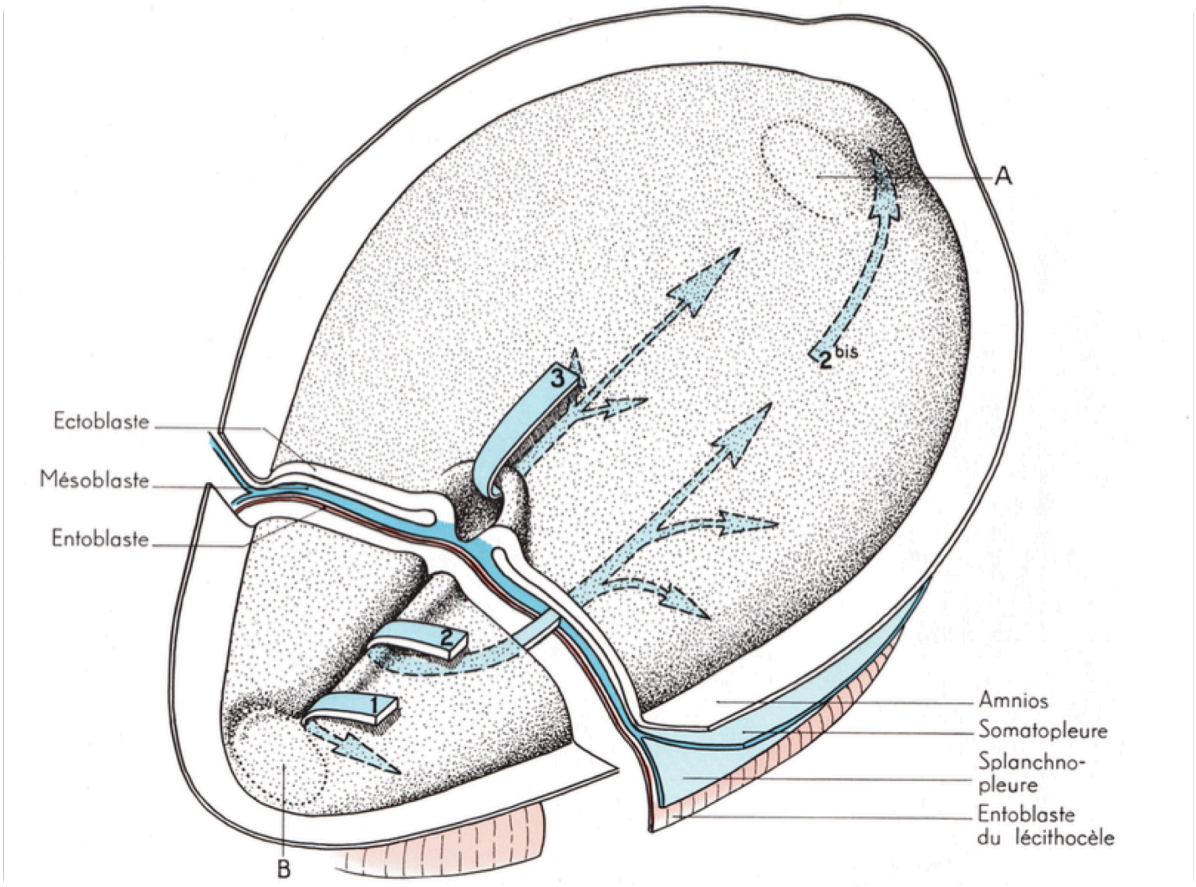
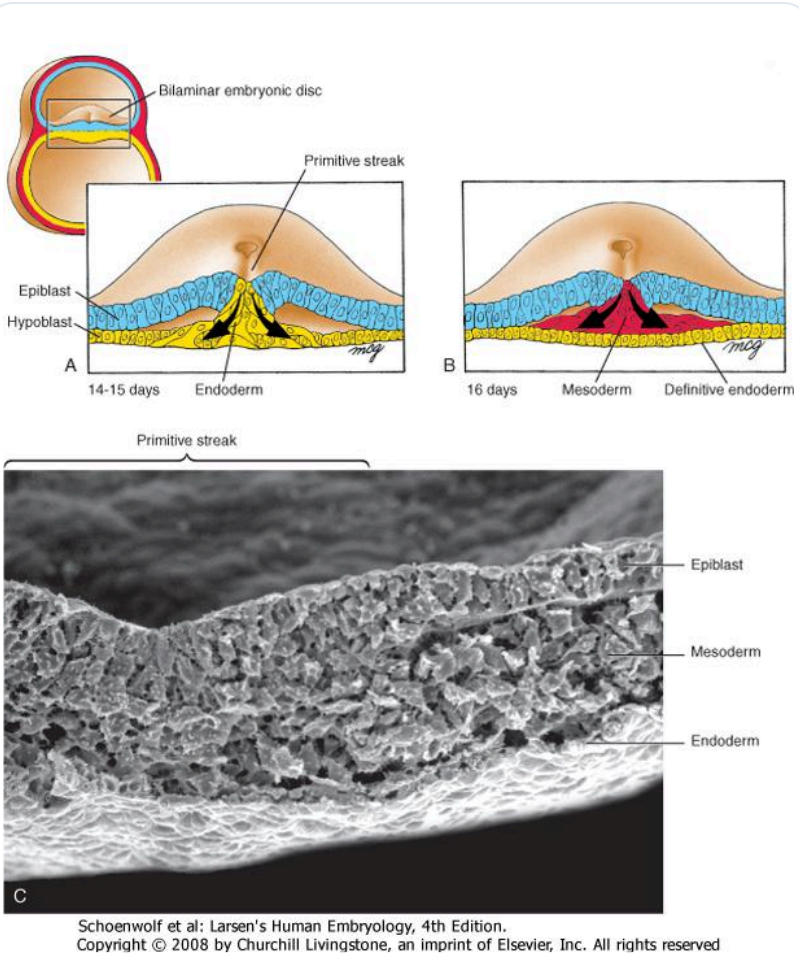


FIG. — Human gastrulation 3 weeks



Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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FIG. — Human gastrulation

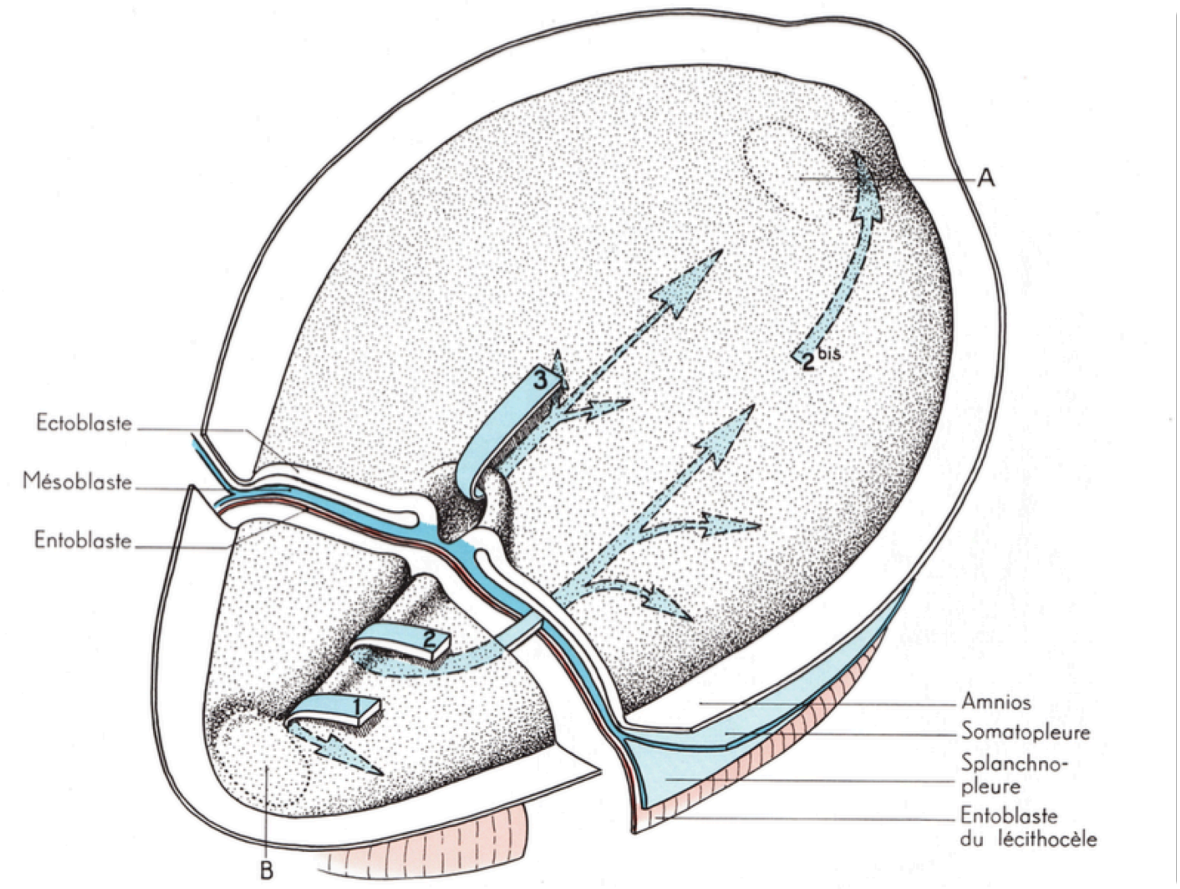


FIG. — Human embryonic gastrulation

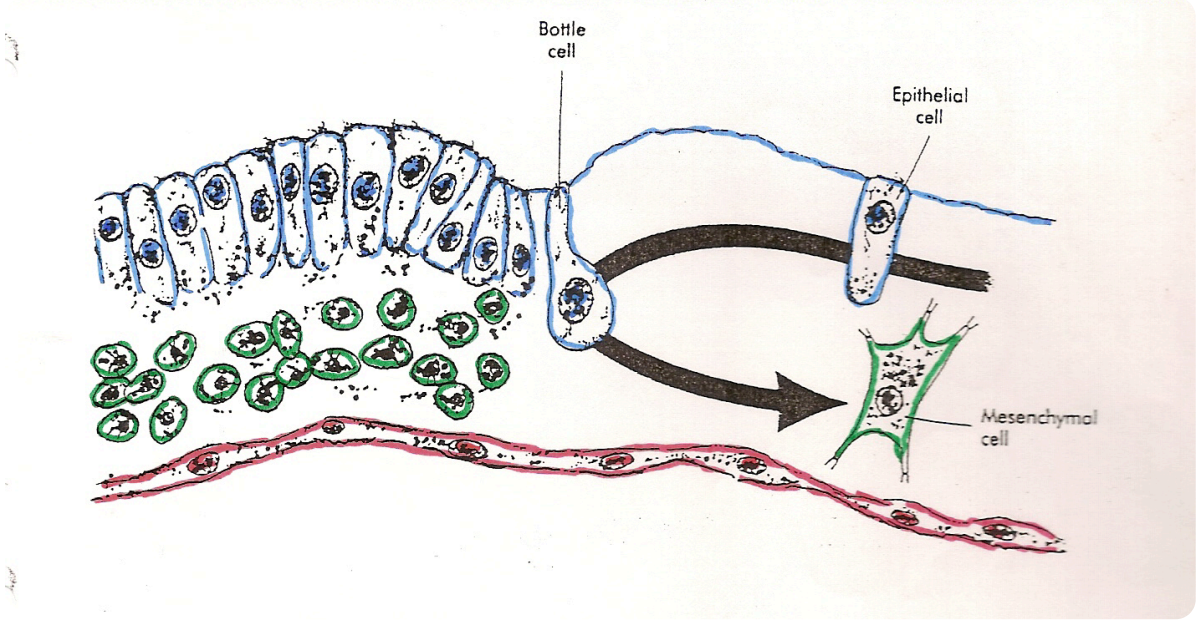


FIG. — Mesenchymal migration

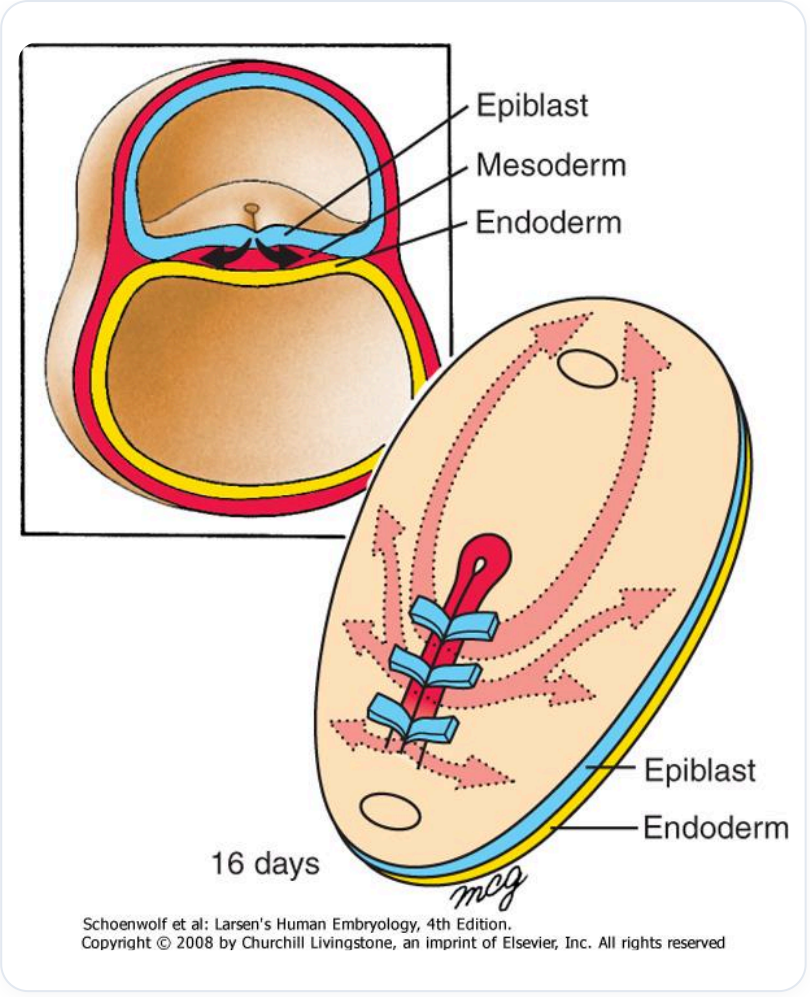


FIG. — Gastrulation 16 days